

GREGORY GOLONKA

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Education

University of Notre Dame

- B.S. Aerospace Engineering, May 2025
 - GPA: 3.53
 - Relevant Coursework: Intermediate Thermodynamics, Fluid Mechanics, Differential Equations I & II, Heat Transfer, Theoretical Aerodynamics, Compressible Aerodynamics, Aerospace Structures, Aerospace Dynamics, Gas Turbines and Propulsions, Flight Mechanics and Intro to Design, Mechanisms and Machines, Orbital and Space Mechanics, Senior Design
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Technical Skills

- Technical & Testing Skills: Mechanical testing, materials characterization, load frame setup, data acquisition, strain analysis, LabVIEW (exposure)
 - Programming & Simulation: Python, MATLAB, Fortran, C++, HTML, Next.js, LabView (exposure)
 - CAD & Design: SolidWorks, Autodesk Fusion 360, XFLR5, XFOIL
 - Tools: CHEMKIN-II, LSOE, ONELAB/Gmsh, Wind Tunnel Instrumentation
 - Engineering: Propulsion Systems, Flight Dynamics, Experimental Aerodynamics
 - Soft Skills: Technical documentation, Leadership, Project Management, Problem Solving, Independent Learning, Collaboration, Adaptability and Flexibility, Communication, Work Ethic, FMEA, RCCA
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Projects & Research

Senior Design — Radio-Controlled Aircraft

2024 - 2025

- Designed a radio-controlled airplane to complete a specific mission, focusing on aerodynamic optimization and structural integrity using finite element analysis (FEA).
- Utilized XFLR5 and SOLIDWORKS CFD to analyze and refine the base model through iterative simulations.
- Led budgeting, logistics, and construction; managed CAD-guided assembly and sourced materials, balancing cost, weight, and durability.

Undergraduate Research — Shock Wave Structure Analysis

2024 - 2025

- Conducted independent study into viscous shock structures under Dr. Joseph M. Powers.
- Investigated dissociation of diatomic molecules under extreme shock conditions using 1D Navier-Stokes equations.
- Advanced understanding of propulsion and detonation modeling by linking molecular dynamics to continuum-scale physics.

Computing Project — Heat Transfer in Food Systems

2025

- Developed a numerical model to simulate transient heat conduction in a steak during cooking.
- Accounted for variable oven temperatures to optimize char formation while maintaining sub-medium-rare interior.
- Created predictive model for cook-time estimation under uncertain parameters.

Personal Project — Portfolio Website -- <https://gtgolonka.com/>

2024 - 2025

- Built first personal website using JavaScript, HTML, Next.js, and CSS.
- Strengthened skills in frontend development and project self-learning.

Gas Turbines & Propulsion — Williams F-107 Engine Redesign

2025

- Performed thermodynamic and combustion cycle analysis for the Williams F-107 turbofan, optimizing efficiency and evaluating combustor performance.
- Analyzed high-temperature fluid behavior and dissociation effects relevant to gas turbine combustion dynamics.
- Applied numerical and analytical modeling to assess combustion system operability, emissions, and thermal loading.

Experimental Aerodynamics — Wind Tunnel Studies

2024

- Calibrated pressure transducers and collected digital data from analog signals.
 - Analyzed airfoil circulation, pressure distribution, and lift/drag characteristics of finite wings.
 - Developed strong foundation in experimental methods and aerodynamic force analysis.
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Work Experience

Movie Theater Projectionist — Capitol Theater (2023–Present)

- Troubleshoot and maintain digital projection systems to ensure uptime of events.

Dorm Front Desk Worker — Dunne Hall, University of Notre Dame (2022–2023)

- Managed front desk operations and supported in-dorm services, ensuring resident needs and safety were met.

Landscape Technician — C. Michaud Landscaping (2021–2024)

- Coordinated team operations across 150+ weekly properties, maintaining high service quality.